Bank Loan Case Study – Final Project-2

Project Description

This project analyzes bank loan application data to identify patterns that indicate whether a customer will face payment difficulties. The aim is to help the company make better loan approval decisions and mitigate risk. The dataset includes application_data.csv (current applications) and previous_application.csv (historical applications). The target variable (TARGET) is 1 if the client faced payment difficulties, otherwise 0.

Approach

- Data quality checks: missingness profiling and imputation (median for numeric, mode for categorical, exclude columns >60% missing).

- Outlier detection: Interquartile Range (IQR) method, robust winsorization at 1st/99th percentiles.

- Class imbalance assessment on TARGET with baseline ratio.

- Exploratory Data Analysis (EDA): univariate distributions, segmented bivariate analysis, engineered ratios.

- Correlation analysis: top numeric drivers vs TARGET, segmentation by contract type, linkage with previous application behavior.

Tech-Stack Used

- Microsoft Excel 2022 (formulas, pivot tables, charts).

- Python 3 with pandas, matplotlib, and python-pptx for reproducibility and automation.

- Files: application_data.csv, previous_application.csv, columns_description.csv.

Insights

- TARGET variable is imbalanced: majority/minority ratio ~ {imbalance_ratio}:1 (class 0 vs 1).

- Top correlations with TARGET include affordability ratios (e.g., CREDIT_INCOME_RATIO, ANNUITY_INCOME_RATIO).

- Age and employment stability (AGE_YEARS, EMPLOY_YEARS) are important drivers of default risk.

- Historical application behavior (PREV_APP_CNT, approval/refusal rates) shows predictive power.

- Segmented analysis reveals contract type impacts correlation strength.

Result

The project produced a cleaned and winsorized dataset, identified key drivers of loan default, and delivered outputs in Excel, PowerPoint, and PDF. The insights are actionable for mitigating loan risks and improving approval processes.

Drive Link

Please insert your Google Drive links here (set visibility to public): - Google Drive (PPT/PDF): - Google Sheet / Excel file: